



Data Mining & Machine Learning – Final Project

Spring 2026

Project Title

Customer Churn Prediction Using Machine Learning

1. Project Overview

Customer churn occurs when a customer stops using a company's services. For telecommunications companies, predicting which customers are likely to leave is extremely important because retaining existing customers is usually cheaper than acquiring new ones.

In this project, students will build a **machine learning system that predicts whether a customer will leave a telecommunications company** based on customer demographic information, account details, and service usage patterns.

Students will apply the **complete data mining pipeline**, including data preprocessing, sampling, feature analysis, classification, and clustering. Multiple machine learning models must be implemented and compared.

2. Learning Objectives

By completing this project, students will:

- Understand how to analyze real-world datasets
- Apply data cleaning and preprocessing techniques
- Handle sampling and dataset splitting
- Study the effect of feature dimensionality
- Implement classification algorithms

- Use clustering to discover hidden patterns
- Evaluate and compare machine learning models

3. Dataset

Students must use the **Telco Customer Churn Dataset**.

Dataset download page:

Telco Customer Churn Dataset

Download link: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Dataset Description

The dataset contains information about **telecommunication customers and their service usage**.

It includes approximately:

- **7043 customer records**
- **21 attributes**

Example attributes include:

- customer ID
- gender
- senior citizen status
- tenure
- internet service
- contract type
- payment method
- monthly charges
- total charges

Target Variable: Churn

Possible values:

Yes → customer left the company

No → customer stayed

The goal of the project is to **predict the churn status of customers**.

4. Project Tasks

Students must complete the following stages.

Stage 1 — Data Understanding

Students must analyze and describe the dataset.

Tasks include:

- explain dataset features
- identify attribute types
- analyze class distribution
- visualize important variables

Possible visualizations:

- histograms
- box plots
- correlation matrix
- feature distributions

Stage 2 — Data Cleaning

Students must preprocess the dataset by:

- handling missing values
- removing inconsistencies
- converting categorical variables into numerical format
- scaling numerical features if necessary

Students must clearly explain their preprocessing steps.

Stage 3 — Sampling

Students must divide the dataset into training and testing sets.

Example split:

- 70% training
- 30% testing

Students should also analyze whether the dataset is **class balanced or imbalanced**.

Stage 4 — Curse of Dimensionality Analysis

Students must study the effect of features on model performance.

Possible tasks include:

- removing redundant attributes
- correlation analysis
- feature importance analysis
- reducing irrelevant features

Students must discuss how feature dimensionality affects results.

Stage 5 — Classification

Students must implement and compare two classification algorithms:

1. Decision Tree Classifier

Students must train a decision tree model to predict churn.

They should analyze:

- tree structure
- feature importance

2. Support Vector Machine (SVM)

Students must train an SVM classifier and compare its performance with the decision tree.

Stage 6 — Model Evaluation

Students must evaluate classification models using:

- accuracy
- precision
- recall
- F1-score
- confusion matrix

Students must compare results between models.

Example comparison table:

MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE
DECISION TREE				
SVM				

5. Implementation Requirements

Students may use:

- Python
- scikit-learn

or

- MATLAB

Students must clearly explain the algorithms and parameters used.

6. Deliverables

Each team must submit the following.

1. Source Code

Well-structured and documented code.

2. Final Report (8–12 pages)

The report must include:

1. Introduction
2. Dataset description
3. Data preprocessing
4. Sampling strategy
5. Feature analysis
6. Model implementation
7. Experimental results

8. Model comparison
9. Discussion
10. Conclusion

3. Presentation

Each team must give a **10–15 minute presentation** explaining:

- dataset
- methods used
- experimental results
- insights discovered

7. Evaluation Criteria

Component	Weight
Data preprocessing	20%
Model implementation	25%
Experimental analysis	25%
Report quality	20%
Presentation	10%

8. Academic Integrity

Students must complete the project independently within their teams.

Copying code or results from other teams or external sources without citation is prohibited.

9. Project Due Date: 12-6-2026

Dr Bashar Tahayna